

# ONE-TO-ONE MAPPINGS ONTO THE CANTOR SET

By M. K. FORT, Jr.\*

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LET  $X$  be a separable metric space. We consider the following conditions on  $X$ :

- ( $\alpha$ )  $X$  is zero dimensional,
- ( $\beta$ ) each point of  $X$  is a limit point of  $X$ ,
- ( $\gamma$ )  $X$  is compact,
- ( $\gamma'$ )  $X$  is a  $G_\delta$  subset of some complete metric space  $M$ .

(We note that ( $\gamma'$ ) is true for a topologically complete space  $X$ . See, for example, [1], p. 207.)

It is well known that if  $X$  satisfies ( $\alpha$ ), ( $\beta$ ) and ( $\gamma$ ), then  $X$  is homeomorphic to the Cantor middle-third set. The following theorem generalizes this result.

Since the proof of our theorem utilizes two elementary properties of upper semi-continuous set-valued functions, we first define this concept and state (without proof) two simple lemmas.

DEFINITION. Let  $F$  be a function on  $X$  such that each  $F(x)$  is a non-empty compact subset of a metric space  $Y$ . Then,  $F$  is upper semi-continuous at  $p$  if corresponding to each open set  $U$  of  $Y$  for which  $F(p) \subset U$  there is a neighborhood  $V$  of  $p$  such that if  $x \in V$  then  $F(x) \subset U$ . The function  $F$  is upper semi-continuous on  $X$  if  $F$  is upper semi-continuous at such point of  $X$ .

LEMMA 1. If  $F_1, F_2, F_3, \dots$  is a sequence of upper semi-continuous set-valued functions,  $F_1(x) \supset F_2(x) \supset F_3(x) \supset \dots$  and  $F(x) = \bigcap_{n=1}^{\infty} F_n(x)$  for each  $x \in X$ , then  $F$  is also upper semi-continuous.

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LEMMA 2. *If  $f$  is a function on  $X$  into  $Y$  and for each  $x \in X$ ,  $F(x)$  is the set whose only member is  $f(x)$ , then  $f$  is continuous if and only if  $F$  is upper semi-continuous.*

THEOREM. *If  $X$  satisfies  $(\alpha)$ ,  $(\beta)$  and  $(\gamma')$ , then there exists a one-to-one mapping (continuous function) on  $X$  onto a Cantor set.*

PROOF. Let  $I$  be the Cantor middle-third set formed for the interval  $[0, 1]$ . We define  $I_0$  to be the set whose only member is 0, and for each positive integer  $k$  we define  $I_k = I \cap [2/3^k, 3/3^k]$ . If  $H$  is the Cantor middle-third set for the interval  $[a, b]$ , we let  $\varphi$  be the mapping of  $I$  onto  $H$  for which  $\varphi(x) = a + (b - a)x$ , and define  $\psi(H, k) = \varphi(I_k)$  for each non-negative integer  $k$ .

Since we are assuming that  $X$  satisfies  $(\gamma')$ , there exist sets  $C_n$ , each closed in  $M$ , such that  $X = \bigcap_{n=1}^{\infty} (M - C_n)$ .

Now, let  $G$  be a subset of  $X$  which is open relative to  $X$ , and let  $n$  be a positive integer. We define a  $(G, n)$  sequence of sets to be a sequence  $S_0, S_1, S_2, \dots$  of sets such that:

- (i)  $S_0$  is a set consisting of a single point;
- (ii) for each  $k > 0$ ,  $S_k$  is both open and closed relative to  $X$ , has diameter less than  $1/n$ , and the closure of  $S_k$  in  $M$  is disjoint from  $C_n$ ;
- (iii)  $S_k \cap S_j = \emptyset$  for  $k \neq j$ ; and
- (iv)  $\bigcup_{k=1}^{\infty} S_k = G$ .

It is easy to prove, using basic properties of zero-dimensional sets, that a  $(G, n)$  sequence of sets exists for each set  $G$  open relative to  $X$  and each positive integer  $n$ .

We next define a sequence  $D_1, D_2, D_3, \dots$  of partitions of  $X$  into subsets. First, let  $D_1$  be the members of some  $(X, 1)$  sequence. If  $D_n$  has been defined, we obtain  $D_{n+1}$  by replacing each non-degenerate (i.e. having more than one element) member  $G$  of  $D_n$  by the members of some  $(G, n + 1)$  sequence.

Our next step is to define a sequence  $F_1, F_2, F_3, \dots$  of upper semi-continuous set-valued functions on  $X$  into the set of compact subsets of  $I$ .

Let  $S_0, S_1, S_2, \dots$  be the  $(X, 1)$  sequence chosen in defining  $D_1$ . If  $x \in S_k$ , we define  $F_1(x) = I_k$ .

Suppose that  $F_n$  has been defined. Let  $x \in X$ . If  $F_n(x)$  consists of a single point, we define  $F_{n+1}(x) = F_n(x)$ . Otherwise,  $x \in G$  for some non-degenerate member  $G$  of the decomposition  $D_n$ . We let  $\Sigma_0, \Sigma_1, \Sigma_2, \dots$  be the  $(G, n+1)$  sequence used in defining  $D_{n+1}$ . Then  $x \in \Sigma_k$  for some  $k$ , and we define  $F_{n+1}(x) = \psi(F_n(x), k)$ . It is easy to see that  $F_1(x) \supset F_2(x) \supset F_3(x) \supset \dots$  for each  $x \in X$ .

We wish to prove that  $F_1$  is upper semi-continuous. Let  $S_0, S_1, S_2, \dots$  be the  $(X, 1)$  sequence chosen in defining  $D_1$ . If  $k > 0$ , then  $S_k$  is open and  $F_1$  is constant on  $S_k$ ; hence  $F_1$  is upper semi-continuous at each point of  $S_k$ . Now let  $p$  be the single member of  $S_0$ , and let  $U$  be an open set which contains  $F_1(p) = I_0$ . It is obvious that there exists  $N$  such that  $I_n \subset U$  for all  $n > N$ . The set  $\bigcup_{j=1}^N S_j$  is closed and does not contain  $p$ . Thus, for  $V = X - \bigcup_{j=1}^N S_j$ , it follows that if  $x \in V$  then  $F_1(x) \subset U$ .

It is now possible to prove by induction that  $F_n$  is upper semi-continuous for all  $n$ . Suppose  $n > 0$ ,  $q \in X$ , and  $F_k$  is upper semi-continuous for all  $k < n$ . If  $F_n(q)$  is non-degenerate, then  $F_n$  is constant on some open set containing  $q$  and hence is upper semi-continuous at  $q$ ; if  $F_{n-1}(q)$  contains only one point then  $F_n(q) = F_{n-1}(q)$  and the upper semi-continuity of  $F_n$  at  $q$  follows from the upper semi-continuity of  $F_{n-1}$  at  $q$  and the fact that  $F_n(x) \subset F_{n-1}(x)$  for all  $x \in X$ ; if  $F_n(q)$  contains only one point but  $F_{n-1}(q)$  is non-degenerate, then  $F_n$  is proved upper semi-continuous at  $q$  in the same way that  $F_1$  was proved upper semi-continuous at  $p$  in the preceding paragraph.

It follows from Lemma 1 that the function  $F$  defined by  $F(x) = \bigcap_{n=1}^{\infty} F_n(x)$  is upper semi-continuous on  $X$ .

For each  $x \in X$ , diameter  $F_n(x) < 3^{-n}$ , and hence  $F(x)$  is a set having a single member. We define a function  $f$  on  $X$  into  $I$  by letting  $f(x)$  be the unique member of  $F(x)$ . By Lemma 2, the upper semi-continuity of  $F$  implies that  $f$  is continuous.

Suppose that  $x$  and  $y$  are different members of  $X$ . Then, there exists a positive integer  $n$  such that  $d(x, y) > 1/n$ . It follows that  $x$  and  $y$  are in different members of the partition  $D_n$  and that  $F_n(x) \cap F_n(y) = \phi$ . Thus  $f(x) \neq f(y)$ , and we see that  $f$  is a one-to-one mapping.

We must show, finally, that  $f$  maps  $X$  onto  $I$ . Let  $q \in I$ . If for some  $n$  and  $p$ ,  $F_n(p)$  is the set whose only member is  $q$ , then  $F_k(p)$  is the set whose only member is  $q$  for all  $k > n$  and it follows that  $f(p) = q$ . Otherwise, for each positive integer  $n$  there exists a non-degenerate member  $G_n$  of the partition  $D_n$  such that  $q \in F_n(x)$  for each  $x \in G_n$ . Since  $M$  is complete,  $G_1 \supset G_2 \supset G_3 \cdots$ , and diameter  $G_n < 1/n$ , there exists  $p \in \bigcap_{n=1}^{\infty} (\text{closure } G_n)$ . Since the closure of  $G_n$  is disjoint from  $C_n$  for each  $n$ ,  $p \in X$ . Since  $G_n$  is closed in  $X$  for each  $n$ ,  $p \in G_n$  for each  $n$  and  $q \in F_n(p)$ . Thus  $f(p) = q$ . This proves that  $f$  is a one-to-one mapping on  $X$  onto  $I$ .

As a corollary to our theorem, we obtain the following well-known result.

**COROLLARY 1.** *If the metric space  $X$  satisfies  $(\alpha)$ ,  $(\beta)$ , and  $(\gamma)$ , then  $X$  is homeomorphic to the Cantor set.*

**PROOF.** Condition  $(\gamma)$  implies condition  $(\gamma')$ , since a compact metric space is complete and is a  $G_\delta$  subset of itself. Hence, by our theorem, there exists a one-to-one mapping  $f$  of  $X$  onto the Cantor set  $I$ . However, since  $X$  is compact, the one-to-one mapping  $f$  must be a homeomorphism.

Our second corollary gives a special case which furnishes an interesting example of a one-to-one mapping which is not a homeomorphism. This result was first proved by Sierpinski (see [2]).

COROLLARY 2. *There exists a one-to-one mapping of the set of all irrational numbers onto the Cantor set.*

PROOF. The set of all irrational numbers satisfies the hypotheses of our theorem, the space  $M$  being the real number system.

The spaces defined in the following three examples each satisfy two of the conditions  $(\alpha)$ ,  $(\beta)$ ,  $(\gamma')$ , but none of them can be mapped in a one-to-one and continuous manner onto the Cantor set.

EXAMPLE 1. The closed interval  $[0, 1]$  satisfies  $(\beta)$  and  $(\gamma')$ , but not  $(\alpha)$ .

EXAMPLE 2. The set whose members are 0 and the reciprocals of the positive integers satisfy  $(\alpha)$  and  $(\gamma')$  (in fact also  $(\gamma)$ ), but not  $(\beta)$ .

EXAMPLE 3. The set of all rational numbers satisfies  $(\alpha)$  and  $(\beta)$ , but not  $(\gamma')$ .

#### REFERENCES

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The University of Georgia  
Athens, Ga. U. S. A.